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Methods of forming nanostructures including metal oxides using block copolymer materials

Abstract

A self-assembled nanostructure comprises first domains and second domains. The first domains comprise a first block of a block copolymer material and an activatable catalyst. The second domains comprise a second block and substantially without the activatable catalyst. The activatable catalyst is capable of generating catalyst upon application of activation energy, and the generated catalyst is capable of reacting with a metal oxide precursor to provide a metal oxide. A semiconductor structure comprises such self-assembled nanostructure on a substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 16/049,329, filed Jul. 30, 2018, now U.S. Pat. No. 11,532,477, issued Dec. 20, 2022, which is a divisional of U.S. patent application Ser. No. 14/920,018, filed Oct. 22, 2015, now U.S. Pat. No. 10,049,874, issued Aug. 14, 2018, which is a divisional of U.S. patent application Ser. No. 14/040,245, filed Sep. 27, 2013, now U.S. Pat. No. 9,177,795, issued Nov. 3, 2015, the disclosure of each of which is hereby incorporated herein in its entirety by this reference.

FIELD

(1) The present disclosure, in various embodiments, relates generally to methods of preparing

nanostructures from self-assembled block copolymer materials selectively permeated with metal oxides and to semiconductor structures including such nanostructures.

BACKGROUND

(2) A continuing goal of integrated circuit fabrication is to decrease the dimensions thereof.

Integrated circuit dimensions can be decreased by reducing the dimensions and spacing of the constituent features or structures. For example, by decreasing the dimensions and spacing of features (e.g., storage capacitors, access transistors, access lines) of a memory device, the overall dimensions of the memory device may be decreased while maintaining or increasing the storage capacity of the memory device.

(3) As the dimensions and spacing of semiconductor device features become smaller, conventional lithographic processes become increasingly more difficult and expensive to conduct. Therefore, significant challenges are encountered in the fabrication of nanostructures, particularly structures having a feature dimension (e.g., critical dimension) of less than a resolution limit of conventional photolithography techniques (about 50 nm). It is possible to fabricate semiconductor structures of such feature dimensions using a conventional lithographic process, such as shadow mask lithography and e-beam lithography. However, use of such processes is limited because the exposure tools are extremely expensive or extremely slow and, further, may not be amenable to formation of structures having dimensions of less than 50 nm.

(4) The development of new lithographic processes, as well as materials useful in such processes, is of increasing importance to make the fabrication of small-scale devices easier, less expensive, and more versatile. One example of a method of fabricating small-scale devices that addresses some of the drawbacks of conventional lithographic techniques is self-assembled block copolymer lithography.

(5) In self-assembled block copolymer lithography, block copolymer materials formed on a substrate are subject to microphase segregation, such as by annealing, to self-assemble the block copolymer materials into ordered nano-scale domains of one block of the block copolymer materials in a matrix of the other block of the block copolymer materials. Then, the domains of one block in the self-assembled block copolymer materials may be selectively removed, leaving the domains of the other block as an etch mask on the substrate. Dimensions of the self-assembled domains so formed are conventionally in the range of 5 nm to 50 nm, which are the dimensions that are extremely difficult to define using conventional lithographic techniques. Therefore, self-assembled block copolymer materials are useful as an etch mask in fabrication of nano-scale semiconductor devices.

(6) Although the self-assembled block copolymer lithography is useful for fabrication of semiconductor structures having dimensions of less than 50 nm, there are still problems that must be addressed. Self-assembled block copolymer structures including the self-assembled block copolymer materials may not provide nano-scale etch masks with sufficient pattern fidelity in terms of line edge and roughness. For example, the self-assembled block copolymer structures often include undesirable random curvatures at the interfaces of domains, resulting in nano-scale etch masks having undesirably high degree of edge roughness and inconsistent widths. One conventional approach to address this problem is by removing (i.e., cutting, chopping) the portions of self-assembled block copolymer domains having undesirable random curvatures, prior to further processing the self-assembled block copolymer structures into nano-scale etch masks.

(7) Additionally, the different domains of the self-assembling block copolymer structures often have little or no etch selectivity from one another. Therefore, improving etch selectivity of the self-assembled domains is desirable.

(8) Accordingly, there is a need for more reliable and less expensive fabrication techniques which are suitable for fabricating complex devices with the desired enhanced density to meet future demands.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1-6 are isometric views of various stages in the preparation of nanostructures from self-assembled block copolymer compositions selectively permeated with metal oxides according to an embodiment of the present disclosure;
- (2) FIGS. 7-12A and 12B are isometric views of various stages in the preparation of nanostructures from self-assembled block copolymer compositions selectively permeated with metal oxides according to another embodiment of the present disclosure; and
- (3) FIGS. 13-17A and 17B are isometric views of various stages in the preparation of nanostructures from self-assembled block copolymer compositions selectively permeated with metal oxides according to yet another embodiment of the present disclosure.

DETAILED DESCRIPTION

(4) The following description provides specific details, such as material types, material thicknesses, and processing conditions in order to provide a thorough description of embodiments of the disclosure. However, a person of ordinary skill in the art will understand that embodiments of the present disclosure may be practiced without employing these specific details. Indeed, the embodiments of the present disclosure may be practiced in conjunction with conventional fabrication techniques employed in the industry.

(5) In addition, the description provided herein does not form a complete process flow for forming a semiconductor device structure, and the semiconductor device structures described below do not form a complete semiconductor device. Only those process acts and structures necessary to understand the embodiments of the present disclosure are described in detail below. Additional acts to form the complete semiconductor device may be performed by conventional fabrication techniques. Also, the drawings accompanying the application are for illustrative purposes only, and are thus not necessarily drawn to scale. Elements common between figures may retain the same numerical designation. Furthermore, while the materials described and illustrated herein may be formed as layers, the materials are not limited thereto and may be formed in other three-dimensional configurations.

(6) As used herein, any relational terms, such as “first,” “second” and “third,” or “top,” “middle” and “bottom,” are used for clarity and convenience in understanding the present disclosure and accompanying drawings and does not connote or depend on any specific preference, orientation or order. It is understood that, although the terms “first,” “second,” “third,” “top,” “middle” and “bottom” are used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

(7) As used herein, the term “substantially,” in reference to a given parameter, property or condition, means to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as within acceptable manufacturing tolerances.

(8) As used herein, the term “substrate” means and includes a base material or construction upon which additional materials are formed. The substrate may be, for example, a semiconductor substrate, a base semiconductor material on a supporting structure, a metal electrode or a semiconductor substrate having one or more materials, structures or regions formed thereon. The substrate may be a conventional silicon substrate, or other bulk substrate comprising a layer of semiconductive material. As used herein, the term “bulk substrate” means and includes not only silicon wafers, but also silicon-on-insulator (SOI) substrates, silicon-on-sapphire (SOS) substrates and silicon-on-glass (SOG) substrates, epitaxial layers of silicon on a base semiconductor foundation, or other semiconductor or optoelectronic materials, such as silicon-germanium (Si.sub.1-xGe.sub.x, where x is, for example, a mole fraction between 0.2 and 0.8), germanium

(Ge), gallium arsenide (GaAs), gallium nitride (GaN), or indium phosphide (InP), among others. Furthermore, when reference is made to a “substrate” in the following description, previous process acts may have been conducted to form materials, regions, or junctions in the base semiconductor structure or foundation. In one embodiment, the substrate is a silicon-containing material, such as a silicon substrate. The substrate may be doped or undoped.

(9) As used herein, the terms “polymer block” or “block” mean and include a grouping of multiple monomer units of a single type (i.e., a homopolymer block) or multiple types (i.e., a copolymer block) of constitutional units into a continuous polymer chain of a desired length that forms part of a larger polymer of an even greater length.

(10) As used herein, the term “block copolymer” means and includes a polymer composed of chains where each chain contains two or more polymer blocks as defined above and at least two of the blocks are of sufficient segregation strength for those blocks to phase separate. A wide variety of block copolymers are contemplated herein including diblock copolymers (i.e., polymers including two polymer blocks), triblock copolymers (i.e., polymers including three polymer blocks), multiblock copolymers (i.e., polymers including more than three polymer blocks), or combinations thereof.

(11) As used herein, the term “amphiphilic block copolymer” means and includes a block copolymer material as defined above with the two or more polymer blocks having different solubilities (i.e., at least one hydrophilic block and at least one hydrophobic block) in water.

(12) The terms “microphase segregation” or “microphase separation,” as used herein, mean and include the property by which homogeneous blocks of a block copolymer material aggregate mutually, and heterogeneous blocks of a block copolymer material separate into distinct domains.

(13) The terms “annealing” or “anneal,” as used herein, mean and include treatment of the block copolymer material so as to enable sufficient microphase separation between two or more different blocks of the block copolymer material to self-assemble into an ordered pattern defined by repeating structural units formed from the blocks.

(14) The term “preferential wetting,” as used herein, means and includes wetting of a block copolymer material wherein the contacting surface at an interface wets one block of the block copolymer material more easily than the other block(s).

(15) The term “neutral wetting,” as used herein, means and includes wetting of a block copolymer material wherein the contacting surface at an interface wets all blocks of the block copolymer material substantially equally.

(16) The term “activatable catalyst,” as used herein, means and includes a material that functions as catalyst upon application of an activation energy, such as photo or thermal energy, and that does not function as a catalyst unless activated.

(17) FIGS. 1-6 show various stages in the preparation of nanostructures from self-assembled block copolymer material selectively permeated with metal oxides according to an embodiment of the present disclosure, wherein the self-assembling of the block copolymer material is directed through graphoepitaxy technique.

(18) FIG. 1 shows a patterned material **110** including a substrate **101**, spacers **102** on the substrate **101**, and trenches **103** defined by a floor **104** (e.g., surface of the substrate **101**) and sidewalls **105** of spacers **102**. The patterned material **110** may be formed using any conventional techniques and, therefore, is not described in detail herein. Although the patterned material **110** of FIG. 1 shows three trenches, it is understood that the patterned material **110** may include one trench, two trenches, or more than three trenches.

(19) As shown in FIG. 2, the trenches **103** of the patterned material **110** may be substantially filled with a block copolymer composition **200** to provide a semiconductor structure **100**. The block copolymer composition **200** may be formed in the trenches **103** by any suitable technique including, but not limited to, spin-coating, blanket coating, spraying, ink coating, or dip coating. Alternatively, the materials may be grown in situ. Depending on the specific block copolymer

composition **200** to be used, the technique for depositing or growing the material may be selected by a person of ordinary skill in the art. While the block copolymer composition **200** described and illustrated herein may be formed as a layer, the materials are not limited thereto and may be formed in other three-dimensional configurations.

(20) The block copolymer composition **200** may include a block copolymer material and an activatable catalyst having higher affinity for one block of the block copolymer material than the other block(s) of the block copolymer material. As a non-limiting example, the activatable catalyst may be a hydrophilic compound, and therefore has higher affinity to the more hydrophilic block than the less hydrophilic (i.e., less polar or non-polar) block of the block copolymer material.

(21) In some embodiments, the block copolymer material may include a more polar block and a less polar block, wherein the activatable catalyst has a higher affinity for the more polar block of the block copolymer material relative to the less polar block. In some embodiments, the volume fraction of the more polar block and the less polar block in the block copolymer material may be from about 20:80 to about 50:50.

(22) In some embodiments, the block copolymer material may be an amphiphilic block copolymer that includes at least one hydrophilic block and at least one hydrophobic block, wherein the activatable catalyst has a higher affinity for the hydrophilic block of the block copolymer material relative to the hydrophobic block.

(23) By way of non-limiting examples, the block copolymer material may include poly(styrene)-b-poly(methylmethacrylate) (PS-b-PMMA), poly(styrene)-b-polyacrylate (PS-b-PA), poly(styrene)-b-poly(methacrylate) (PS-b-PMA), poly(styrene)-b-poly(vinylpyridine) (PS-b-PVP), poly(styrene)-b-poly(lactide) (PS-b-PLA), poly(styrene)-b-poly(tert-butyl acrylate) (PS-b-PtBA), poly(styrene)-b-poly(ethylene-co-butylene (PS-b-(PS-co-PB))), poly(styrene)-b-poly(ethyleneoxide) (PS-b-PEO), poly(isoprene)-b-poly(ethyleneoxide) (PI-b-PEO), poly(isoprene)-b-poly(methylmethacrylate) (PI-b-PMMA), poly(butadiene)-b-poly(ethyleneoxide) (PBD-b-PEO), polystyrene-block-polydimethylsiloxane (PS-b-PDMS), polyethyleneoxide-b-polymethylmethacrylate (PEO-b-PMMA), polystyrene-b-poly(9,9-bis(6'-N,N,N-trimethylammonium)hexyl)-fluorene phenylene) (PS-b-PFP), polyethylethylene-b-polyethyleneoxide (PEE-b-PEO), polystyrene-b-polyisoprene (PS-b-PI), polystyrene-b-polybutadiene (PS-b-PBD), polystyrene-b-polyferrocenyldimethylsilane (PS-b-PFS), polybutadiene-b-polyvinylpyridine (PBD-b-PVP), polyisoprene-b-polymethylmethacrylate (PI-b-PMMA), poly(styrene)-b-PEO block copolymer having a cleavable junction between PS and PEO blocks, poly(styrene)-b-poly(methylmethacrylate) doped with PEO-coated gold nanoparticles, poly(styrene)-b-poly(2-vinylpyridine) (PS-b-P2VP) block copolymer having a cleavable junction, poly(styrene-b-methyl methacrylate-b-ethylene oxide) (PS-b-PMMA-b-PEO), poly(styrene-b-methyl methacrylate-b-styrene) (PS-PMMA-PS), poly(methyl methacrylate-b-styrene-b-methyl methacrylate) (PMMA-PS-PMMA), poly(styrene-b-poly(isoprene)-b-styrene) (PS-b-PI-b-PS), or combinations thereof.

(24) The activatable catalyst may be activatable acid catalyst, activatable base catalyst, or a mixture thereof. By way of non-limiting examples, the activatable acid catalyst may be photo acid generator (PAG), thermal acid generator (TAG), or a mixture thereof. By way of non-limiting examples, the activatable base catalyst may be photo base generator (PBG), thermal base generator (TBG), or a mixture thereof. Non-limiting examples of PAGs may include iodonium sulfonate compounds, sulfonium sulfonate compounds, sulfonium triflate compounds, iodonium triflate compounds, iodonium phosphate compounds, imide sulfonate compounds, triazine compounds, diazodisulfone compounds, or a mixture thereof. Non-limiting examples of TAGs may include alkyl ester compounds, sulfonic ester compounds, organic sulfonic acid compounds, blocked alkyl phosphoric acid compounds, blocked perfluoralkyl sulfonic acid compounds, alkyl phosphoric acid/amine complexes, amine compounds, quaternary ammonium compounds, or combinations thereof. Non-limiting examples of PBGs may include sulfonamide compounds, carbamate compounds, O-acyloxime compounds, nifedipine compounds, formamide compounds, ammonium salts,

amineimide compounds, alpha-aminoketone compounds, aromatic urea compounds, metallocene compounds, transition metal complexes, or a mixture thereof. Non-limiting examples of TBGs may include acyloxyimino compounds.

(25) As the activatable catalyst has higher affinity for one block of the block copolymer material than the other block(s), the block copolymer material, upon exposure to conditions that cause self-assembly of the block copolymer composition, may include activatable catalyst selectively in one block of the copolymer material and not in the other block(s).

(26) The amount of activatable catalyst in the block copolymer composition may be varied depending on various factors, including, but are not limited to, the morphologies of the self-assembled domains of the block copolymer material, the chemical structures of the more polar blocks, the chemical structure of the activatable catalyst, the processing time and temperature, etc. As discussed in detailed hereinafter, the different morphologies of self-assembled domains have different volume fractions of the more and less polar blocks, and therefore different morphologies of self-assembled domains require different amounts of the activatable catalyst. In some embodiments, the amount of the activatable catalyst may not be more than about 20% by weight of the block copolymer composition. When the amount of the activatable catalyst in the composition is too high, the activatable catalyst may affix to the less polar block of the block copolymer material in addition to the more polar block, resulting in poor selectivity of the activatable catalyst for the desired domains of the self-assembled structure, and consequently poor fidelity of the nanostructures. On the contrary, when the amount of the activatable catalyst in the composition is too low, the formation of metal oxide in the self-assembled structure may be unacceptable low due to insufficient reaction of generated catalyst and the metal oxide precursor within reasonable processing conditions.

(27) The block copolymer composition **200** in the trenches **103** may be subject to microphase separation to self-assemble the block copolymer composition into domains of a first block of the block copolymer material and domains of a second block, wherein the first block is more polar than the second block and includes the activatable catalyst. Microphase separation of the block copolymer composition may be achieved using any conventional process and therefore is not described in detail herein. In some embodiments, the block copolymer composition may be subject to annealing process to cause microphase separation. For way of non-limiting examples, the annealing process may include thermal annealing, solvent vapor-assisted annealing, supercritical fluid-assisted annealing, photo annealing, or a combination thereof.

(28) Referring to FIG. 3, upon microphase separation, the block copolymer composition **200** self-assembles into domains **201** composed of a first block and an activatable catalyst in a matrix **202** composed of a second block of the block copolymer material, wherein the first block is more polar than the second block. In some embodiments, as shown in FIG. 3, the more polar block is a minor block and forms domains **201** within the matrix **202** of the less polar block, and the self-assembled structure has the activatable catalyst selectively in the minor domains **201** and not in the matrix **202**. Alternatively, in some embodiments, the more polar block may be a majority block and form a matrix for domains of the less polar block, and the self-assembled structure has the activatable catalyst selectively in the matrix of the more polar block and not in the minor domains of the less polar block.

(29) In some embodiments, as shown in FIGS. 1-6, the microphase separation of the block copolymer composition to form self-assembled nano-scale structures may be achieved by graphoepitaxy technique. The floor **104** and sidewalls **105** of the trench **103** guide self-assembly of the blocks of the block copolymer composition **200** to form self-assembled domains **201** including at least one polymer block within a matrix **202** of at least one different polymer block.

(30) In some embodiments, the self-assembled nano-scale structures of the block copolymer composition may be achieved by chemoepitaxy technique, which will be discussed in FIGS. 13-17. In yet further embodiments, a combination of chemically and topographically patterned substrates

may be used to direct the self-assembly of the block copolymer composition.

(31) Various size and shape of the self-assembled domains may be formed from microphase separation of the block copolymer composition. By way of non-limiting examples, the self-assembled domains may have lamellar, cylindrical, half cylindrical, or spherical morphologies, among others. The size and shape of the self-assembled domains formed from microphase separation of the block copolymer composition may be controlled by various factors depending on the characteristics of the block copolymer material in the composition, as well as the wetting ability of the interface surfaces with the block copolymer composition. The molecular weight and volume fraction of the different polymer blocks of the block copolymer material may contribute to the size and shape of the self-assembled domains. For example, for volume fractions at ratios greater than about 80:20 of the two blocks (A:B) of a diblock AB copolymer, the block copolymer composition may microphase separate and self-assemble into spherical domains with spheres of block B surrounded by a matrix of block A. For volume fractions of the two blocks (A:B) between about 60:40 and about 80:20, the block copolymer composition may self-assemble into a single array or hexagonal array of cylinders of block B within a matrix of block A. For volume fractions of the two blocks (A:B) between about 50:50 and 60:40, alternating lamellar domains of the A and B blocks may be formed. In some embodiments, the size of self-assembled domains may be from about 5 nm to about 50 nm. Furthermore, the morphologies of the self-assembled block copolymer composition may be directed by the wetting abilities of the interface surfaces (e.g., the floor and sidewalls of the trench). As a non-limiting example, when the block copolymer composition is formed within the trench having a neutral wetting floor and preferentially wetting sidewalls, the block copolymer composition may self-assemble into alternating lamellar domains oriented perpendicular to the floor of the trench and parallel to the sidewalls, or cylindrical domains in a single array or hexagonal array perpendicular to the floor of the trench and parallel to the sidewalls. As a non-limiting example, when the block copolymer composition is formed within the trench having preferentially wetting floor and sidewalls, the block copolymer composition may self-assemble into half cylindrical domains oriented parallel to both the floor and sidewalls of the trench.

(32) In FIG. 4, selected portions of the semiconductor structure **100** (i.e., area below line **300-300** in direction of “X”) may be subjected to activation energy to generate catalyst from the activatable catalyst in the selected portions. Thus, the self-assembled structure in the selected portions may include catalyst-containing domains **301** composed of the first block and the generated catalyst, and domains **302** composed of the second block of the block copolymer material. The domains **302** may be substantially the same as the matrix **202**, since there is substantially no activatable catalyst in the matrix **202**, and, consequently, no generated catalyst in the domains **302**. The activatable catalyst in the non-activated portions of structure **100** is not subject to activation energy, and therefore no catalyst is generated from the activatable catalyst in the domains **201** of the non-activated portions of structure **100**. Non-limiting examples of activation energy may include photo energy or thermal energy. The activation energy may be applied to only the selected portions of the semiconductor structure **100** using any conventional technique, which is not described in detail herein. By way of non-limiting examples, the technique may be a masking technique, direct application technique (i.e., without using mask), or a combination thereof. The selection of activation energy may be based on various factors including, but not limited to, the type of activatable catalyst (e.g., thermal or photo generated catalyst), the chemical structure of activatable catalyst, or the desirable rate of activation.

(33) FIG. 5 shows the semiconductor structure **100** obtained after exposing the semiconductor structure **100** of FIG. 4 to a metal oxide precursor. The generated catalyst (e.g., acid or base) may react with the metal oxide precursor through a condensation reaction, resulting in a formation of metal oxide. Thus, the generated catalyst in the catalyst-containing domains **301** may react with the metal oxide precursor through a condensation reaction to form metal oxide, resulting in metal

oxide-containing domains **400** that include the first block of the block copolymer material and metal oxide. As the domains **201**, **302** and matrix **202** of the semiconductor structure **100** do not have the generated catalyst, no metal oxide is formed in these domains. Therefore, the metal oxide is selectively incorporated at the selected portions of the semiconductor structure **100** and at the selected time, by controlling the location and time for the application of the activation energy to the semiconductor structure **100**.

(34) Various known metal oxide precursors may be used. Non-limiting examples of metal oxide precursors may include metal alkoxide, metal chloride, metal β -diketonate, etc. A variety of metal oxides may be selectively permeated in the self-assembled structure. Non-limiting examples of metal for metal oxides may include titanium (Ti), aluminum (Al), zirconium (Zr), hafnium (Hf), or silicon (Si). The metal oxide precursor may be in gas phase, liquid phase, or solid form. In some embodiments, the metal oxide precursor is in gas phase.

(35) In some embodiments, as shown in FIG. 6, the domains **201**, **302** and matrix **202** of the semiconductor structure **100** of FIG. 5 may be removed, leaving the domain **400** that includes the first block and the metal oxide on the substrate **101**. Since the domains **201**, **302** and matrix **202** do not include the metal oxide, the domains **201**, **302** and matrix **202** may be selectively removed using any conventional process, which is not described in detail herein. Thus, by incorporating the metal oxide into a portion of the block copolymer material, the etch selectivity of the block copolymer material may be increased.

(36) A method of forming nano structures may comprise forming a block copolymer composition within a trench in a material on a substrate, wherein the block copolymer composition may comprise a block copolymer material and an activatable catalyst having a higher affinity for a first block of the block copolymer material compared to a second block of the block copolymer material; self-assembling the block copolymer composition into first domains comprising the first block and the activatable catalyst, and second domains comprising the second block; generating catalyst from the activatable catalyst in at least one portion of the first domains to produce a structure comprising catalyst-containing domains and the second domains, the catalyst-containing domains comprising the first block and the catalyst; and reacting a metal oxide precursor with the catalyst in the catalyst-containing domains to produce a metal oxide-containing structure comprising the first block and metal oxide.

(37) A self-assembled nanostructure may comprise first domains and second domains, wherein the first domains may comprise a first block of a block copolymer material and an activatable catalyst, and the second domains may comprise a second block of the block copolymer material and substantially without the activatable catalyst, and wherein the activatable catalyst is capable of generating catalyst upon application of activation energy, the catalyst capable of reacting with a metal oxide precursor to provide a metal oxide.

(38) As discussed above, the formation of the metal oxide at certain portions of the self-assembled block copolymer structure may be accomplished by applying the activation energy to the selected portions of the structure to generate catalyst from the activatable catalyst at the selected portions, and the generated catalyst reacts with the metal oxide precursor to form metal oxide at the selected portions of the self-assembled block copolymer structure.

(39) FIGS. 7-12A and 12B show various stages in the preparation of nanostructures from self-assembled block copolymer materials according to another embodiment of the present disclosure, wherein different metal oxides may be formed at different portions of the self-assembled block copolymer structure.

(40) FIG. 7 shows the semiconductor structure **100** obtained by applying activation energy to the selected portion (i.e., area below the line **300-300** in direction of "X") of the semiconductor structure **100** of FIG. 3 to generate catalyst from the activatable catalyst in the selected portions. As shown in FIG. 7, the selected portions of the self-assembled block copolymer structure include domains **301** composed of the first block and the generated catalyst, and domains **302** composed of

the second block of the block copolymer material. The domains **302** may be substantially the same as the matrix **202**, since there is substantially no activatable catalyst in the matrix **202**. Therefore, the self-assembled block copolymer structure of FIG. 7 may include the generated catalyst selectively in the domains **301**, and not in the domains **201**, **302** and matrix **202**.

(41) FIG. 8 shows the semiconductor structure **100** obtained by exposing the semiconductor structure **100** of FIG. 7 to a first metal oxide precursor. The generated catalyst in the catalyst-containing domains **301** may react with the first metal oxide precursor through a condensation reaction to form the first metal oxide, resulting in metal oxide-containing domains **400** that include the first block of the block copolymer material and the first metal oxide. The semiconductor structure **100** of FIG. 8 having the first metal oxide selectively incorporated in the domains **400**, and not in the domains **201**, **302** and matrix **202** of the semiconductor structure.

(42) FIG. 9 shows the semiconductor structure obtained by applying activation energy to a selected portion (i.e., the area below line **500-500** in direction of “X” but above the line **300-300**) of the semiconductor structure **100** of FIG. 8 to generate catalyst from the activatable catalyst in the selected portions. As shown in FIG. 9, the selected portions of the self-assembled block copolymer structure may include catalyst-containing domains **502** composed of the first block and the generated catalyst, and matrix **202** composed of the second block of the block copolymer material. The matrix **202** of the selected areas may remain substantially unchanged, since there is substantially no activatable catalyst in the matrix **202**. Therefore, the self-assembled block copolymer structure of FIG. 9 may include generated acid or base selectively in the catalyst-containing domains **502**, and not in the domains **201** and matrix **202**.

(43) FIG. 10 shows the semiconductor structure **100** obtained by exposing the semiconductor structure **100** of FIG. 9 to a second metal oxide precursor. The catalyst in the catalyst-containing domains **502** of FIG. 9 may react with the second metal oxide precursor to provide the semiconductor structure **100** of FIG. 10 having the second metal oxide formed selectively in the metal oxide-containing domains **602**, and not in the domains **400**, **201** and matrix **202** of the semiconductor structure.

(44) FIG. 11 shows the semiconductor structure **100** having domains **700** composed of a third metal oxide. Upon application of activation energy to the semiconductor structure **100** of FIG. 10, catalyst may be generated from the activatable catalyst in the domains **201** of FIG. 10. Consequently, upon exposure to a third metal oxide precursor, the catalyst in the domains **201** may react with the third metal oxide precursor to form the third metal oxide selectively in the area above line **600-600** in the direction of “X” of the domains **700** of the semiconductor structure **100** of FIG. 11.

(45) In some embodiments, as shown in FIG. 12A, the matrix **202** of the semiconductor structure **100** of FIG. 11 may be removed, leaving the domains **400** including the first block and the first metal oxide, the domains **602** including the first block and the second metal oxide, and the domains **700** including the first block and the third metal oxide on the substrate **101**. The matrix **202** may be removed using any conventional process, and therefore is not described in detail herein.

(46) In some embodiments, as shown in FIG. 12B, the block copolymer material in the domains **400**, **602**, **700** and matrix **202** in the semiconductor structure **100** of FIG. 11 may be removed, leaving ordered nanostructures of the first metal oxide **402**, the second metal oxide **604**, and third metal oxide **702** on the substrate **101**. The block copolymer material in the domains **400**, **602**, **700** and matrix **202** may be removed using any conventional process, and therefore is not described in detail herein.

(47) A semiconductor structure may comprise a self-assembled nanostructure on a substrate, the self-assembled nanostructure comprising self-assembled domains in a matrix, wherein the self-assembled domains may comprise at least a first portion and a second portion, the first portion comprising a first block of a block copolymer material and a first metal oxide, the second portion comprising the first block of the block copolymer material and a second metal oxide, and wherein

the matrix may comprise a second block of the block copolymer material and substantially without any metal oxide.

(48) Although FIGS. **1-12B** show that the block copolymer composition self-assembles through the graphoepitaxy technique, it is understood that the block copolymer composition may self-assemble through the chemoepitaxy technique or other techniques. Furthermore, although FIGS. **1-12B** show the nanostructures in straight line features, it is understood that the nanostructures may have any other desired structures. Non-limiting examples of other structures may include zig-zag lines, curved lines, or a 1-3-1 line pattern.

(49) FIGS. **13-17A** and **17B** show various stages in the preparation of nanostructures from self-assembled block copolymer material selectively permeated with metal oxides according to an embodiment of the present disclosure, wherein the block copolymer material self-assembles through chemoepitaxy technique.

(50) FIG. **13** shows a chemically patterned material **110'** that includes a substrate **101'**, portions **102'a**, and portions **102'b**, wherein the portions **102'a** are chemically different from the portions **102'b**. The portions **102'b** may have greater affinity to at least one polymer block of the block copolymer composition and be used to direct the self-assembly of the block copolymer composition.

(51) In FIG. **14**, the block copolymer composition **200'** is formed on the portions **102'a**, **102'b** of the patterned material **110'**. The block copolymer composition **200'** includes a block copolymer material and an activatable catalyst having a higher affinity to a more polar block of the block copolymer material than a less polar block of the block copolymer material. The portions **102'b** of the patterned material **110'** have chemical specificity to the more polar blocks of the block copolymer composition; whereas, the portions **102'a** of patterned material **110'** have chemical specificity to the less polar blocks of the block copolymer composition.

(52) Upon subjecting a semiconductor structure **100'**, of FIG. **14**, to microphase separation, the portions **102'a**, **102'b** of the patterned material **110'** direct the chemoepitaxy self-assembly of the block copolymer composition **200'** to provide the semiconductor structure **100'** of FIG. **15**. The structure of FIG. **15** includes domains **202'a** over the portions **102'a** of patterned material **110'**, and domains **202'b** over the portions **102'b** of patterned material **110'**. The domains **202'b** may include the more polar block and the activatable catalyst. The domains **202'a** may include the less polar block of the block copolymer material, and are substantially lacking the activatable catalyst.

(53) FIG. **16** shows a semiconductor structure **100'** obtained by applying activation energy to selected portions of the semiconductor structure **100'** of FIG. **15** to generate catalyst from the activatable catalyst in the selected portions. The semiconductor structure **100'** of FIG. **16** may include domains **301'** composed of the more polar block and the catalyst generated from the activatable catalyst, and domains **202'a** composed of the less polar block of the block copolymer material. The domains **202'a** remain unchanged after the application of activation energy, since there is substantially no activatable catalyst in the domains **202'a**. The activatable catalyst in the non-activated portions (i.e., not subjected to the activation energy) remains unactivated, and therefore there is no catalyst generated in the domains **202'b**. The activation energy may be selectively applied to the desired portions of the semiconductor structure **100'** using any conventional technique, and therefore is not described in detail herein.

(54) The semiconductor structure **100'** of FIG. **16** may be exposed to a metal oxide precursor to provide the semiconductor structure **100'** of FIG. **17A**. The catalyst in the domain **301'** may react with the metal oxide precursor to form self-assembled block copolymer domains **400'** composed of the more polar block of the block copolymer material and metal oxide. There is substantially no metal oxide formed in domains **202'a** and **202'b**, since the domains **202'a** have substantially no activatable catalyst, and the domains **202'b** have substantially no generated catalyst.

(55) A method of forming nano structures may comprise forming a block copolymer composition on a patterned material, wherein the block copolymer composition may comprise a block

copolymer material and an activatable catalyst having higher affinity for a first block of the block copolymer material compared to a second block; self-assembling the block copolymer composition into a structure comprising domains of the first block and the activatable catalyst in a matrix of the second block; applying an activation energy to a first portion of the domains of the first block and the activatable catalyst to produce a catalyst in the first portion of the domains; and exposing the structure to a first metal oxide precursor to form a first metal oxide in the first portion of the domains.

(56) In some embodiments, the domains **202'a**, **202'b** in the semiconductor structure **100'** of FIG. **17A** may be removed, leaving self-assembled block copolymer domains **400'** composed of the more polar block of the block copolymer material and metal oxide on the patterned material **110'**. The domains **202'a** and **202'b** may be removed using any conventional process, and therefore is not described in detail herein.

(57) In some embodiments, as shown in FIG. **17B**, the block copolymer material in the domains **202'a**, **202'b** and **400'** in the semiconductor structure **100'** of FIG. **17A** may be removed, leaving ordered nanostructures **402'** composed of metal oxide on the patterned material **110'**. The block copolymer material in the domains **202'a**, **202'b** and **400'** may be removed using any conventional process, and therefore is not described in detail herein. The nanostructures **402'** may be subject to further processing steps to form nano-scale features for the semiconductor devices.

(58) The semiconductor structures **100**, **100'** of FIGS. **6**, **12A**, **12B**, **17A** and **17B** may be subjected to further processing for fabrication of a semiconductor device. In some embodiments, the semiconductor structures may be further processed by conventional techniques to form nano-scale etch masks having dimensions of less than 50 nm with improved pattern fidelity (line edge and roughness). In some embodiments, the semiconductor structures, such as those shown in FIGS. **12B** and **17B**, may be further processed by conventional techniques to form nano-scale metal routing or nano-scaled metal features for fabrication of a semiconductor device.

(59) The method of forming nanostructures may provide nano-scale etch masks with improved pattern fidelity in terms of line edge and roughness, and substantially without undesirable random curvature, thereby minimizing, if not eliminating, the need of removing random curvature.

(60) While the present disclosure is susceptible to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and have been described in detail herein. However, the present disclosure is not intended to be limited to the particular forms disclosed. Rather, the present disclosure is to cover all modifications, equivalents, and alternatives falling within the scope of the present disclosure as defined by the following appended claims and their legal equivalents.

Claims

1. A method of forming nanostructures, comprising: forming a block copolymer composition on a patterned material, the block copolymer composition comprising a block copolymer material and an activatable catalyst having higher affinity for a first block of the block copolymer material compared to a second block of the block copolymer material; self-assembling the block copolymer composition by microphase segregation into first domains comprising the first block and the activatable catalyst, and second domains comprising the second block; applying an activation energy to a first portion of the first domains to activate the activatable catalyst in the first portion of the first domains; reacting a first metal oxide precursor with the activated activatable catalyst in the first portion of the first domains to form a first metal oxide in the first portion of the first domains; applying an activation energy to a second portion of the first domains to activate the activatable catalyst in the second portion of the first domains; and reacting a second metal oxide precursor with the activated activatable catalyst in the second portion of the first domains to form a second metal oxide in the second portion of the first domains.

2. The method of claim 1, further comprising: applying an activation energy to a third portion of the first domains to produce a catalyst in the third portion of the first domains; and reacting a third metal oxide precursor with the catalyst in the third portion of the first domains to form a third metal oxide in the third portion of the first domains.
3. The method of claim 2, further comprising removing the second domains to provide a nanostructure comprising the first portion, the second portion, and the third portion of the first domains.
4. The method of claim 3, further comprising removing the first block of the block copolymer material from the first portion, the second portion, and the third portion of the first domains to form a nanostructure comprising the first metal oxide, the second metal oxide, and the third metal oxide on the patterned material.
5. The method of claim 1, wherein applying an activation energy to a first portion of the first domains comprises applying photo energy, thermal energy, or a combination thereof to the first portion of the first domains.
6. The method of claim 1, wherein forming a first metal oxide in the first portion of the first domains comprises forming the first metal oxide comprising a metal selected from the group consisting of titanium (Ti), aluminum (Al), zirconium (Zr), hafnium (Hf), silicon (Si), and combinations thereof and forming a second metal oxide in the second portion of the first domains comprises forming the second metal oxide comprising a metal selected from the group consisting of titanium (Ti), aluminum (Al), zirconium (Zr), hafnium (Hf), silicon (Si), and combinations thereof, the metal of the first metal oxide different than the metal of the second metal oxide.
7. The method of claim 1, wherein forming a block copolymer composition on a patterned material comprises forming the block copolymer material comprising the first and second blocks, the first block being more hydrophilic than the second block.
8. The method of claim 1, wherein forming a block copolymer composition on a patterned material comprises forming the block copolymer material comprising the first and second blocks at a volume ratio of from 20:80 to 50:50.
9. The method of claim 1, wherein reacting a second metal oxide precursor with the activated activatable catalyst in the second portion of the first domains comprises forming the second metal oxide comprising a different metal than the first metal oxide.
10. A method of forming nanostructures, comprising: forming a block copolymer composition on a patterned material, the block copolymer composition comprising a block copolymer material and an activatable catalyst having higher affinity for a first block of the block copolymer material compared to a second block of the block copolymer material; self-assembling the block copolymer composition by microphase segregation into first domains comprising the first block and the activatable catalyst, and second domains comprising the second block; applying an activation energy to a first portion of the first domains to activate the activatable catalyst in the first portion of the first domains and to generate activated activatable catalyst; reacting a first metal oxide precursor with the activated activatable catalyst in the first portion of the first domains to form a first metal oxide in the first portion of the first domains comprising the first block of the block copolymer material; applying an activation energy to a second portion of the first domains to activate the activatable catalyst in the second portion of the first domains and to generate activated activatable catalyst, the second portion of the first domains adjacent to the first portion of the first domains; and reacting a second metal oxide precursor with the activated activatable catalyst in the second portion of the first domains to form a second metal oxide in the second portion of the first domains, the second metal oxide comprising a different metal than the first metal oxide.
11. The method of claim 10, further comprising: applying an activation energy to a third portion of the first domains to activate the activatable catalyst in the third portion of the first domains and to generate activated activatable catalyst; and reacting a third metal oxide precursor with the activated activatable catalyst in the third portion of the first domains to form a third metal oxide in the third

portion of the first domains.

12. The method of claim 11, further comprising removing the second domains to produce a nanostructure comprising the first metal oxide, the second metal oxide, and the third metal oxide.
 13. The method of claim 10, wherein reacting a first metal oxide precursor with the activated activatable catalyst in the first portion of the first domains and reacting a second metal oxide precursor with the activated activatable catalyst in the second portion of the first domains comprises forming substantially no first metal oxide or second metal oxide in the second domains comprising the second block.
 14. The method of claim 10, wherein forming a block copolymer composition on a patterned material, the block copolymer composition comprising a block copolymer material and an activatable catalyst comprises forming the block copolymer composition comprising the block copolymer material and the activatable catalyst selected from the group consisting of a photo acid generator (PAG), a thermal acid generator (TAG), a photo base generator (PBG), a thermal base generator (TBG), and a mixture thereof.
 15. The method of claim 10, wherein self-assembling the block copolymer composition by microphase segregation into first domains comprising the first block and the activatable catalyst, and second domains comprising the second block comprises self-assembling the block copolymer composition by graphoepitaxy or by chemoepitaxy.
 16. The method of claim 10, wherein forming a block copolymer composition on a patterned material comprises forming the block copolymer composition comprising a relatively more polar block and a relatively less polar block on the patterned material.
 17. The method of claim 10, wherein reacting a second metal oxide precursor with the activated activatable catalyst in the second portion of the first domains comprises forming the second metal oxide directly adjacent to the first metal oxide.
 18. The method of claim 10, wherein reacting a second metal oxide precursor with the activated activatable catalyst in the second portion of the first domains comprises forming the second metal oxide in the second portion of the first domains without forming the second metal oxide in the second domains.
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